

GIACOMO GUARESI

Full-Stack Developer

React · TypeScript · Node.js · C# / .NET · PostgreSQL · Docker

Full-stack developer with eight years of shipping production software: React and TypeScript front ends, Node.js and .NET back ends, PostgreSQL, Docker and CI/CD. I built and run the internal platform of a manufacturing startup end to end, and worked on enterprise web applications as a consultant. A background in industrial systems and IoT means I am at home where software meets real-world data. Milan area, Italy — open to hybrid or remote roles.

guaresi.giacomo@gmail.com giacomoguaresi.github.io/GiacomoGuaresi github.com/GiacomoGuaresi
linkedin.com/in/giacomo-guaresi-b76b0812b

SKILLS

Frontend	React · TypeScript · JavaScript · Angular · HTML · CSS
Backend	Node.js · C# / .NET (Framework & Core) · Python · Java / Spring · REST API design · OAuth · Kafka · Redis pub/sub · Monolith, microservices, event-driven
Data	PostgreSQL · MySQL · MongoDB · Oracle · Redis · SQLite · Query tuning and indexing on datasets up to ~1 TB
DevOps & cloud	Docker · Docker Compose · Kubernetes (managed) · GitHub Actions · GitLab CI · Linux servers · Production deploys, domains, TLS · Git
AI	LLM integration with function calling (Gemini) · OpenCV · TensorFlow
Also	Embedded and IoT — C / C++, ESP32, STM32, CAN bus, MQTT, Modbus, OPC-UA, embedded Linux, PCB design

EXPERIENCE

Ginger Additive — Software Engineer (Full-Stack)	2024 — present
- Built and maintain around ten internal web applications the company runs on — project management first among them — with React, Node.js and PostgreSQL. - Own the infrastructure underneath: Linux servers, Docker deployments, CI pipelines, domains and TLS, from first commit to production. - Built an AI assistant inside the project management tool: Gemini with function calling creates and updates tasks and answers questions about projects in plain language. In daily use. - Also responsible for the software of the company's industrial 3D printer: control stack on Klipper and custom Linux images, several PCBs, and upstream contributions to OrcaSlicer, Klipper and Mainsail.	
Intre — Full Stack Developer (consulting)	2023 — 2024
- Front end and back end on enterprise web applications for large industrial and professional-services clients (details under NDA). React, Angular and TypeScript front ends over containerised, cloud-native services and APIs on managed Kubernetes, inside large codebases and multi-team projects.	
AM Instruments — Software Developer, industrial monitoring & R&D	2019 — 2023
- Built the .NET web portal and the iOS and Android apps for a room sterilisation system used in pharmaceutical plants, plus its Python tooling. - Tuned databases and process historians approaching a terabyte (indexing, query optimisation) for monitoring systems with hundreds of sensors.	

- Integrated monitoring software (SCADA: iFIX, Movicon) with plant equipment over Modbus, OPC-UA, Profinet and EtherCAT, under GMP and ISO 9001.
- Applied defect detection, predictive maintenance and adaptive thresholds to process data (OpenCV, TensorFlow).
- Wrote the embedded C++ for the sterilisation controller (ARM, embedded Linux, CAN bus).

Interlem — Full Stack Developer

2018 – 2019

- Built a **Spring** and **PostgreSQL** web portal for managing and reordering a fleet of IoT-connected gas cylinders (LoRa smart scales, hardware built in a team).
- Refactored a legacy Delphi business portal and maintained it until its .NET replacement shipped.

PROJECTS

FAInder

Web app and scraper for the FAI heritage sites across Italy (Python).

Open source

Contributions to OrcaSlicer, Klipper and Mainsail.

EDUCATION & LANGUAGES

Diploma

Perito Informatico (technical diploma in computer science) — ITIS Hensemberger, Monza, 2018

Languages

Italian (native) · English (professional working proficiency)

Learning

Rust · Agentic AI systems